Extra Dessert #16: VHF/UHF Operating Methods

MSK144 meteor scatter, Q65 EME, FT8/FT4 contest exchanges, APRS, AX.25, packet digipeaters.

Group E2D · 11 pool questions · Self-study reference · Practice exams at eham.net/exams

WSJT Modes For Weak-Signal VHF/UHF

Question	Answer	Q
Digital mode designed for meteor scatter	**MSK144** (E2D01). Short, fast bursts during meteor ionization.	E2D01
Digital mode designed for EME	**Q65** (E2D03). Successor to JT65, optimized for deep moonbounce paths.	E2D03
Information that replaces SNR in FT8/FT4 VHF contests	**Grid square** (E2D02). The 4-character Maidenhead locator becomes the exchange.	E2D02
Characteristic of JT65 mode	**Decodes signals with a very low signal-to-noise ratio** (E2D05). Classic weak-signal performer.	E2D05
Method for establishing EME contacts	**Time-synchronous transmissions alternating between stations** (E2D06). Each station transmits during fixed time slots.	E2D06
Modulation used by JT65	**Multitone AFSK** (E2D09). Many tones in a narrow band, encoded for very low SNR decoding.	E2D09

APRS And Packet Networks

Question	Answer	Q
Technology used for real-time balloon tracking	**APRS** (E2D04). Automatic Packet Reporting System with GPS-tagged position beacons.	E2D04
Digital protocol used by APRS	**AX.25** (E2D07). Same packet protocol as classic amateur packet radio.	E2D07
Type of packet frame used for APRS beacon data	**Unnumbered Information** (E2D08). UI frames are connectionless and broadcast to any listener.	E2D08
How APRS stations relay data	**By packet digipeaters** (E2D11). A digipeater hears a beacon and re-transmits it onward.	E2D11
What does the packet path WIDE3-1 designate?	**Three digipeater hops are requested with one remaining** (E2D10). The 3 is the total request; the 1 is what is left.	E2D10

Dessert 16 Question Bank (E2D)

E2D01. Which of the following digital modes is designed for meteor scatter communications?
A. WSPR.
B. MSK144.
C. Hellschreiber.
D. APRS.

E2D02. What information replaces signal-to-noise ratio when using the FT8 or FT4 modes in a VHF contest?
A. RST report.
B. State abbreviation.
C. Serial number.
D. Grid square.

E2D03. Which of the following digital modes is designed for EME communications?
A. MSK144.
B. PACTOR III.
C. WSPR.
D. Q65.

E2D04. What technology is used for real-time tracking of balloons carrying amateur radio transmitters?
A. FT8.
B. Bandwidth compressed LORAN.
C. APRS.
D. PACTOR III.

E2D05. What is the characteristic of the JT65 mode?
A. Uses only a 65 Hz bandwidth.
B. Decodes signals with a very low signal-to-noise ratio.
C. Symbol rate is 65 baud.
D. Permits fast-scan TV transmissions over narrow bandwidth.

E2D06. Which of the following is a method for establishing EME contacts?
A. Time-synchronous transmissions alternating between stations.
B. Storing and forwarding digital messages.
C. Judging optimum transmission times by monitoring beacons reflected from the moon.
D. High-speed CW identification to avoid fading.

E2D07. What digital protocol is used by APRS?
A. PACTOR.
B. QAM.
C. AX.25.
D. AMTOR.

E2D08. What type of packet frame is used to transmit APRS beacon data?
A. Acknowledgement.
B. Burst.
C. Unnumbered Information.
D. Connect.

E2D09. What type of modulation is used by JT65?
A. Multitone AFSK.
B. PSK.
C. RTTY.
D. QAM.

E2D10. What does the packet path WIDE3-1 designate?
A. Three stations are allowed on frequency, one transmitting at a time.
B. Three subcarriers are permitted, subcarrier one is being used.
C. Three digipeater hops are requested with one remaining.
D. Three internet gateway stations may receive one transmission.

E2D11. How do APRS stations relay data?
A. By packet ACK/NAK relay.
B. By C4FM repeaters.
C. By DMR repeaters.
D. By packet digipeaters.

Answer Key

E2D01: B	E2D02: D	E2D03: D	E2D04: C
E2D05: B	E2D06: A	E2D07: C	E2D08: C
E2D09: A	E2D10: C	E2D11: D	